

LLMThings: Experimentation Platform for Language Agents-Based Orchestration of IoT Systems

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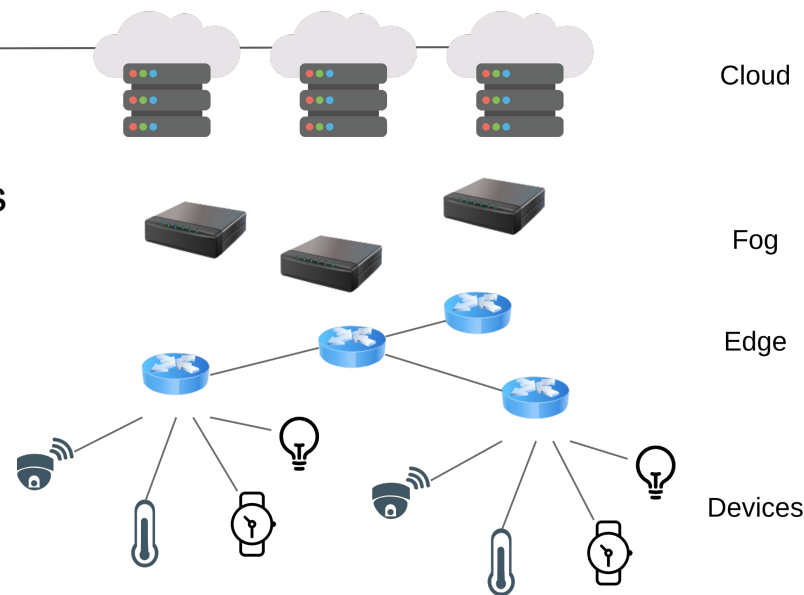
²Laboratoire IBISC, Université Paris-Saclay, Evry, France.



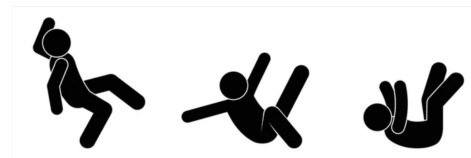
Motivation

Orchestration: **Coordinating IoT services/resources** across the continuum Edge/Fog/Cloud to perform a given task

- (1) Service composition and workflow
- (2) Resource allocation and placement
- (3) Heterogeneity and interoperability
- (4) Runtime adaptation



Problem: IoT deployments are increasingly **heterogeneous, large-scale, and dynamic**, making orchestration difficult [1].



IoT system monitoring patients within the nursing home, e.g., for fall prevention

Related works: Orchestration of IoT Systems

- **Classical approaches:** predefined policies, heuristics, mathematical optimization, distributed coordination [4, 5].
- **Machine learning:** reinforcement learning (e.g., Q-learning, DQN), fuzzy decision trees, neural network-based resource prediction [6, 7].
- **Multi-agent systems:** decentralized collective learning (EPOS Fog), agents locally propose and cooperatively select assignments [8].

Existing approaches very often require **domain expertise**, are **rigid**, and **fail to adapt** to new, unseen situations.

Related works: LLM-based Orchestration

Recent approaches leverage LLMs' reasoning capabilities for flexible, adaptive orchestration able to handle never-seen-before situations.

System	Service Composition	Resource Placement	Heterog. & Interop.	Mobility & Adaptation
WirelessAgent [2]		✓		✓
LLMind 2.0 [3]	✓		✓	✓
Sasha [9]	✓			
AutoloT [10]	✓			
IoT-MCP [11]			✓	

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limited to
a **single agent**

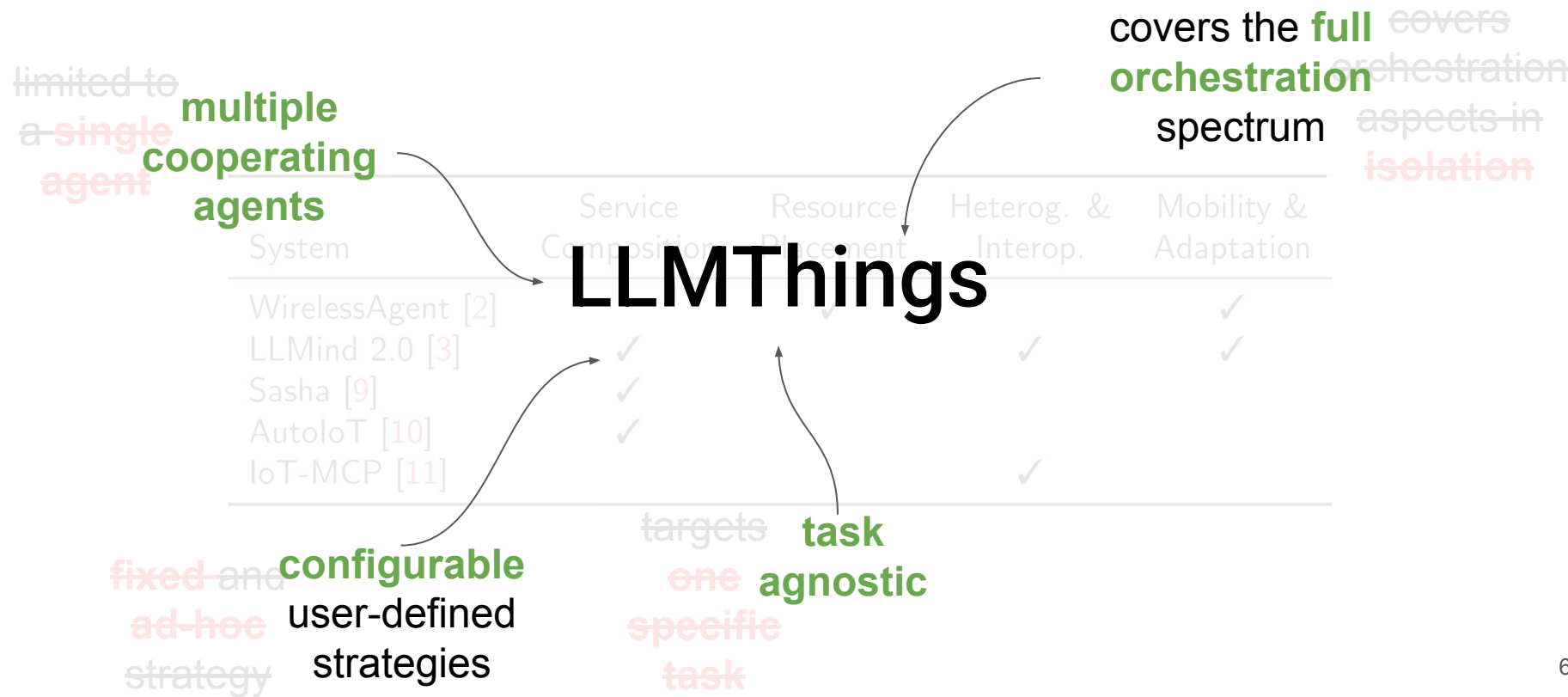
covers
orchestration
aspects in
isolation

System	Service Composition	Resource Placement	Heterog. & Interop.	Mobility & Adaptation
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fixed and ad-hoc strategy

targets
one specific task

LLMThings: Our contribution



LLMThings: Our contribution

LLMThings provides an experimentation platform for developing and evaluating LLM-agent-based orchestration strategies toward autonomic IoT systems.

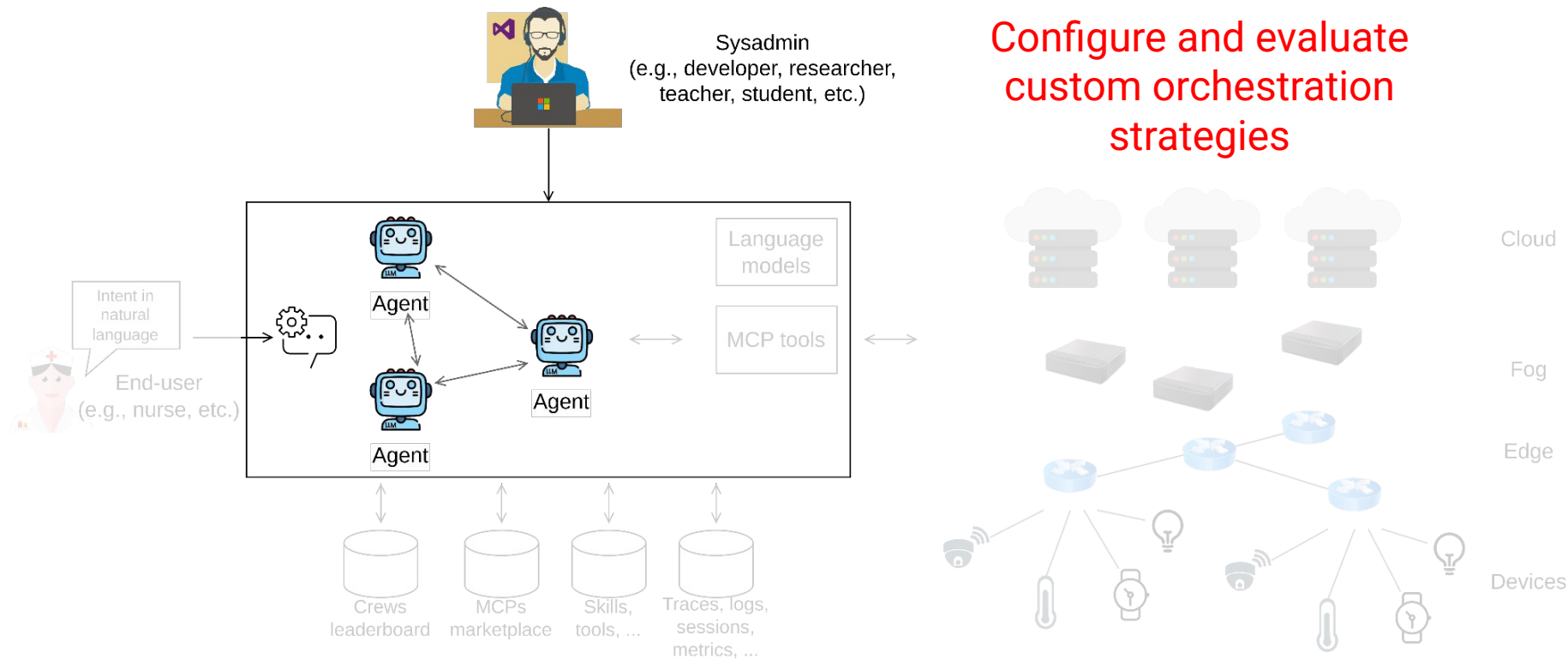
Define

Execute

Observe

Evaluate

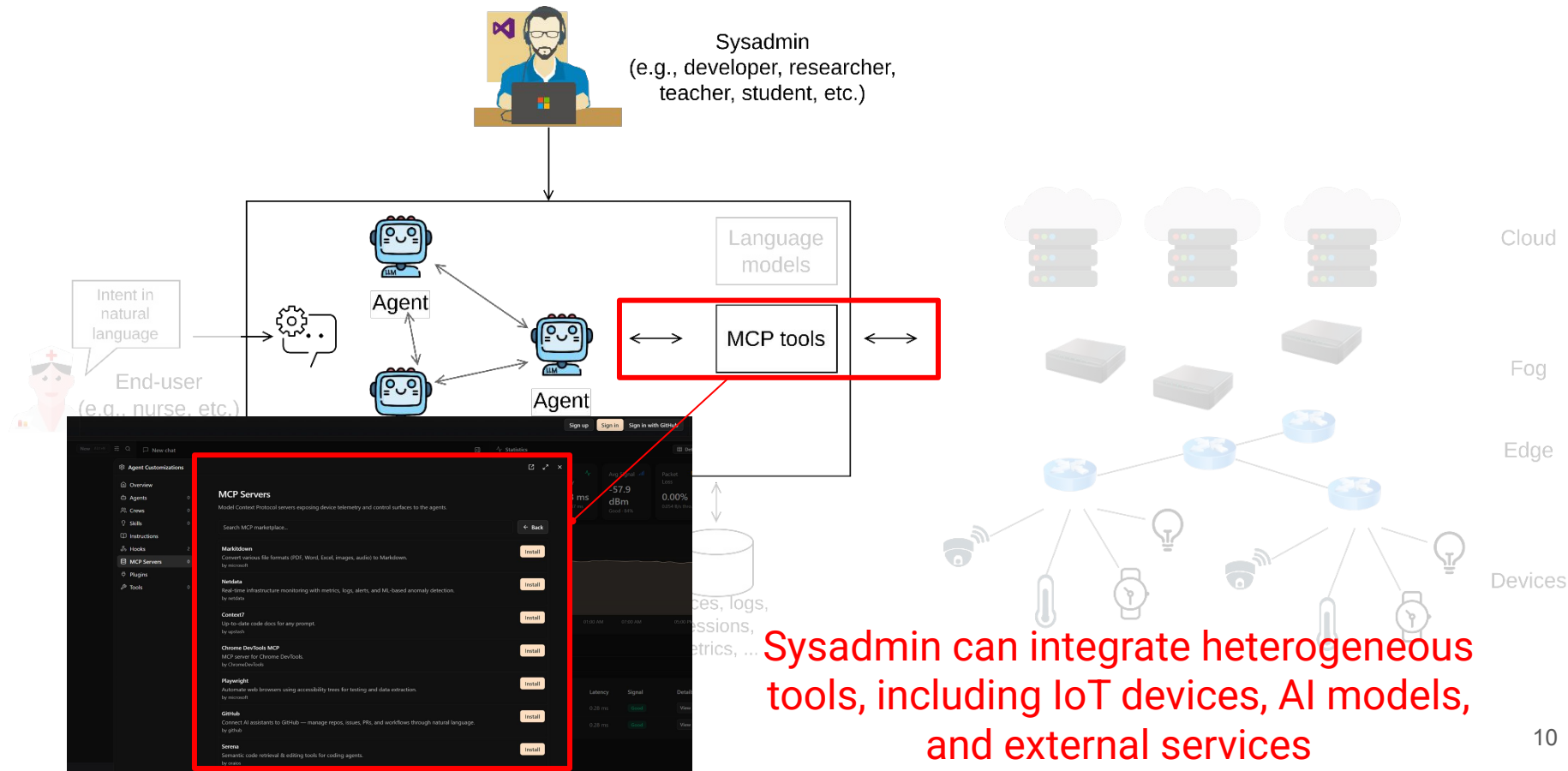
LLMThings: High-Level Architecture



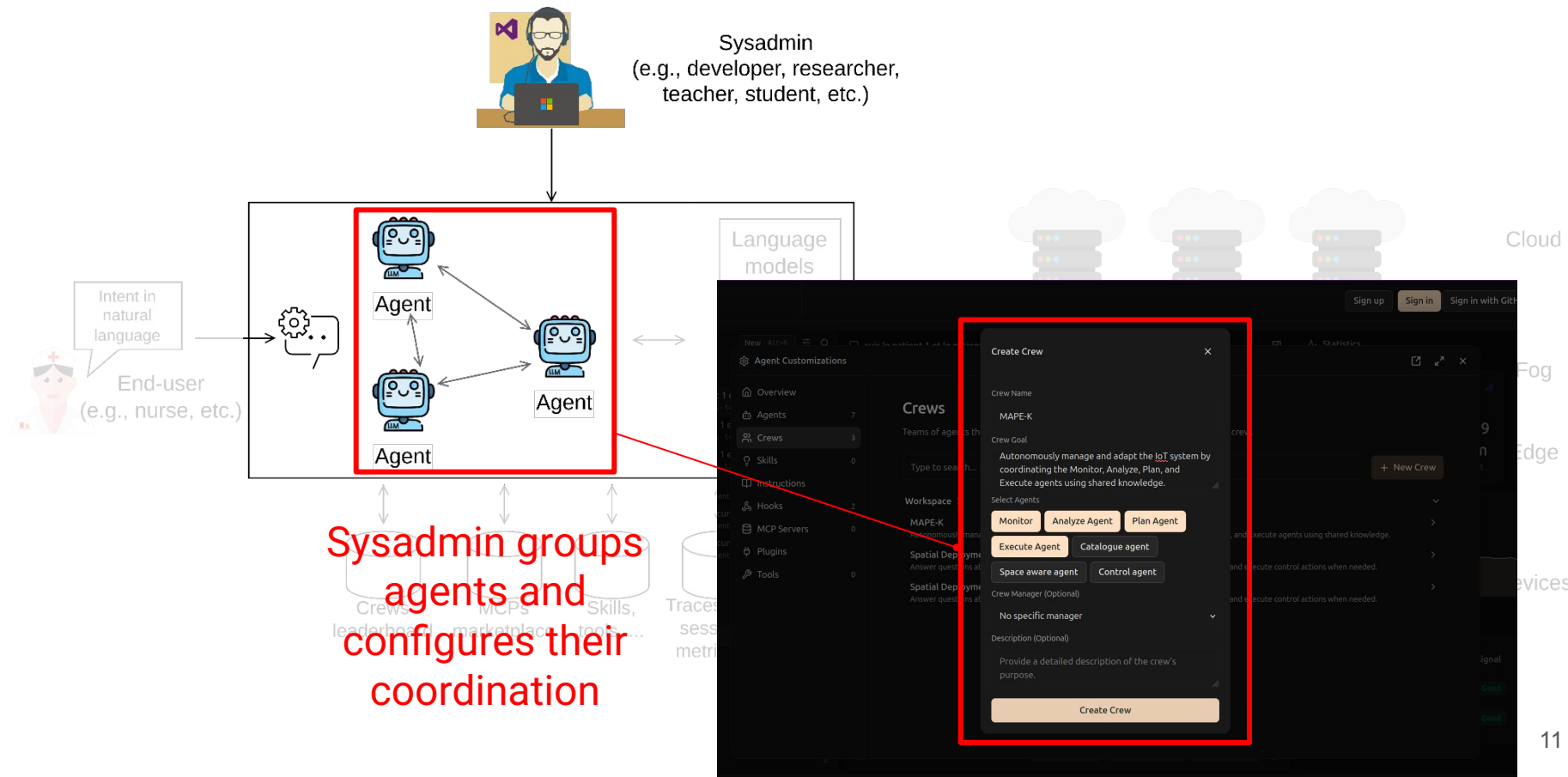
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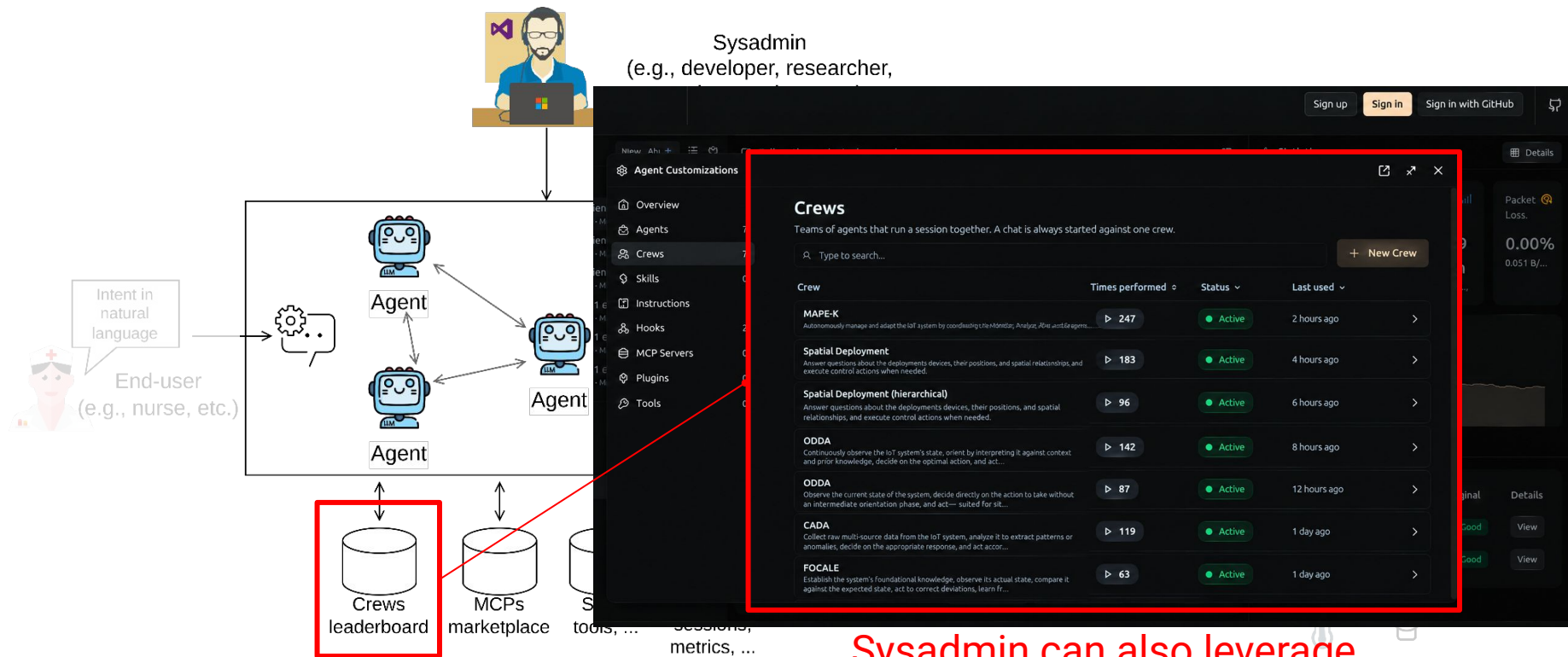
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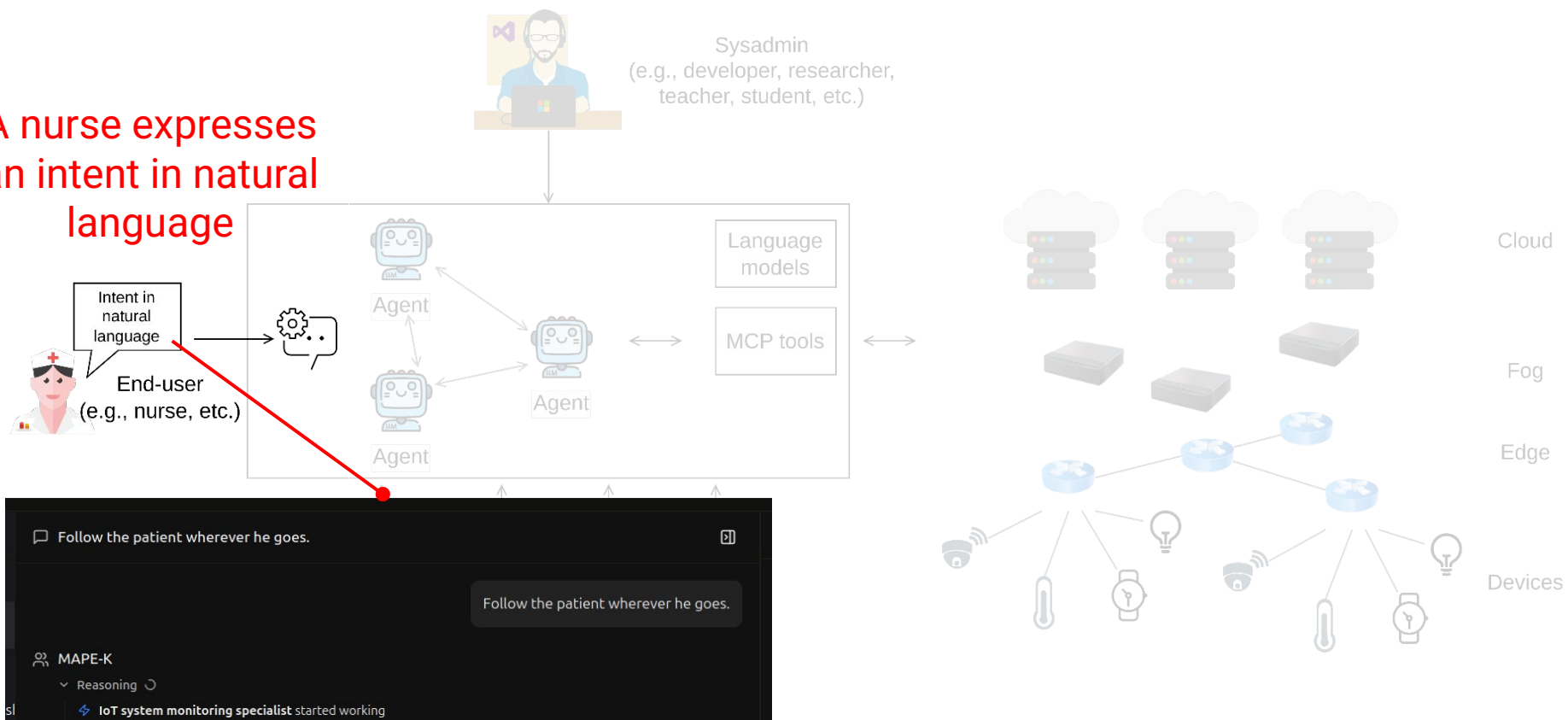
LLMThings: High-Level Architecture



Sysadmin can also leverage pre-existing crews that performed well on certain tasks

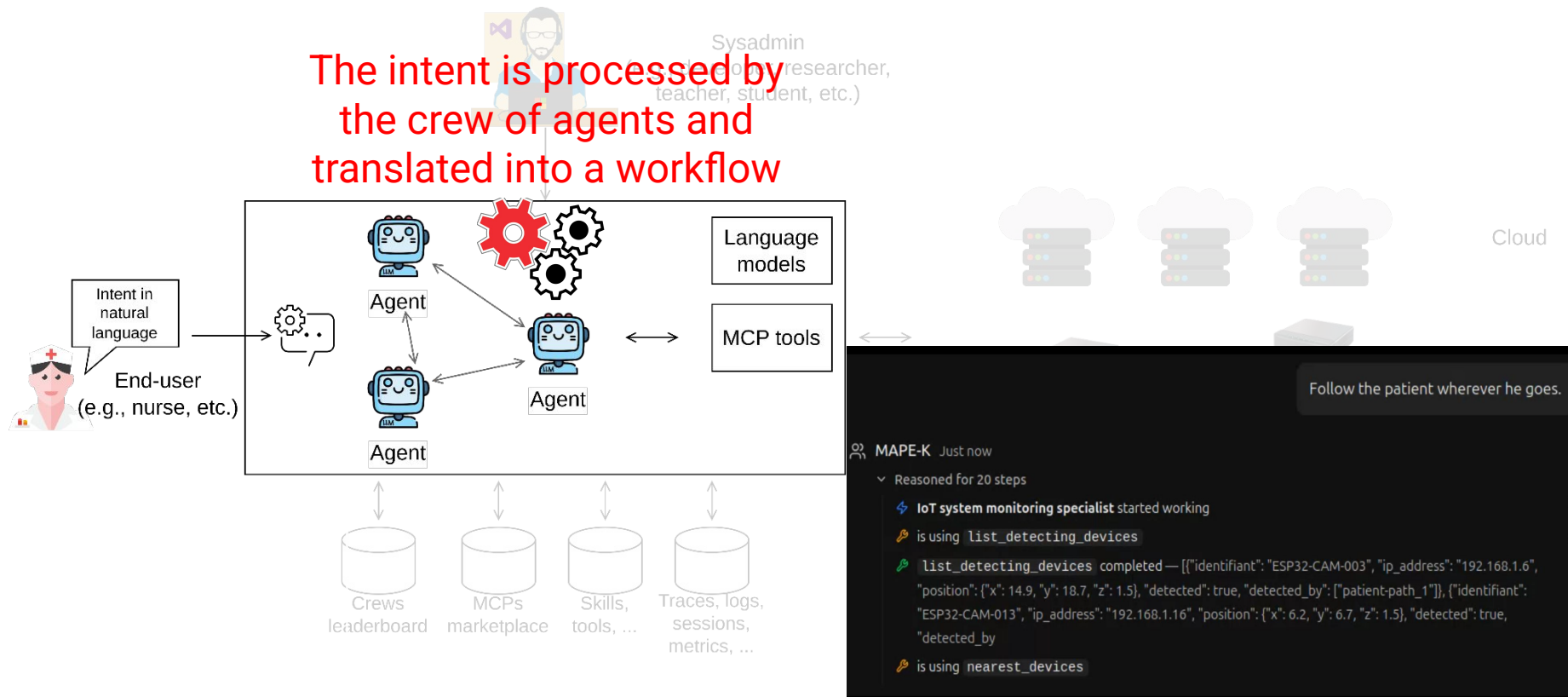
LLMThings: High-Level Architecture

A nurse expresses an intent in natural language

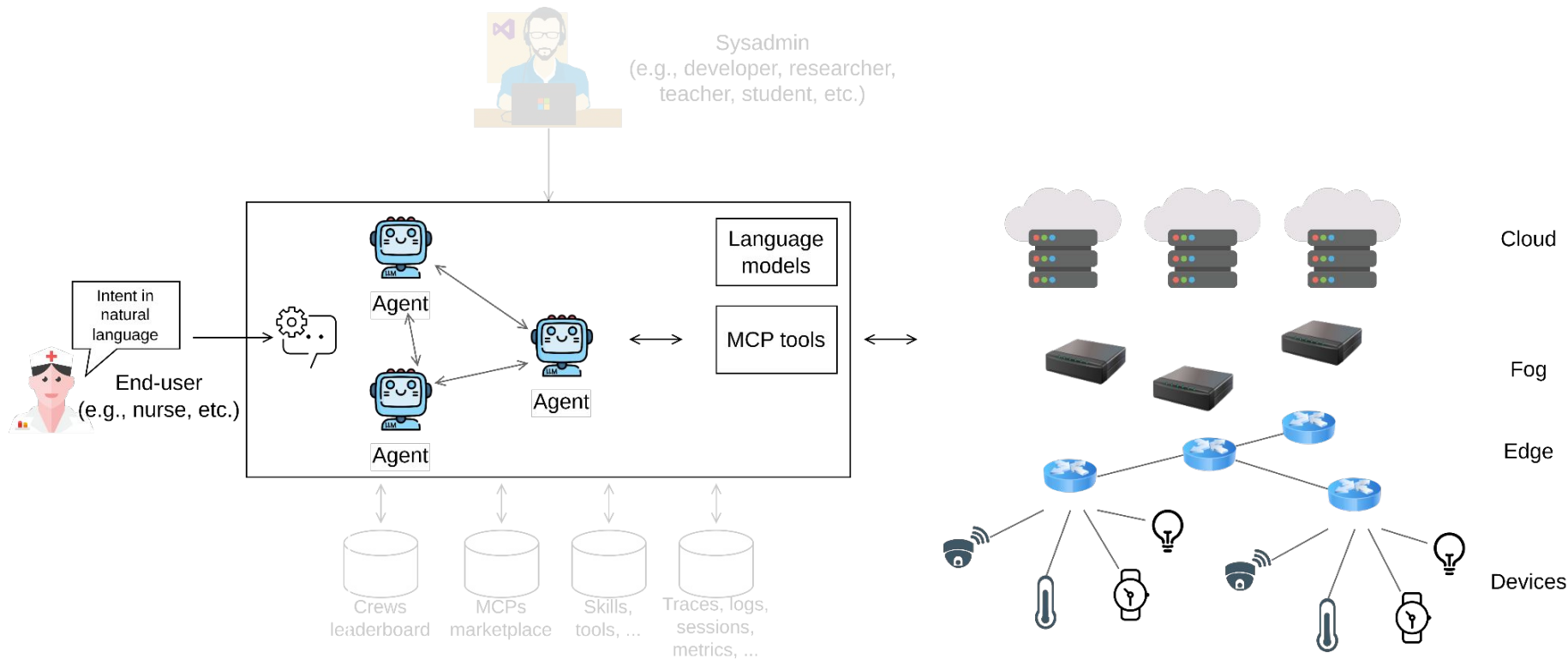


LLMThings: High-Level Architecture

The intent is processed by
the crew of agents and
translated into a workflow

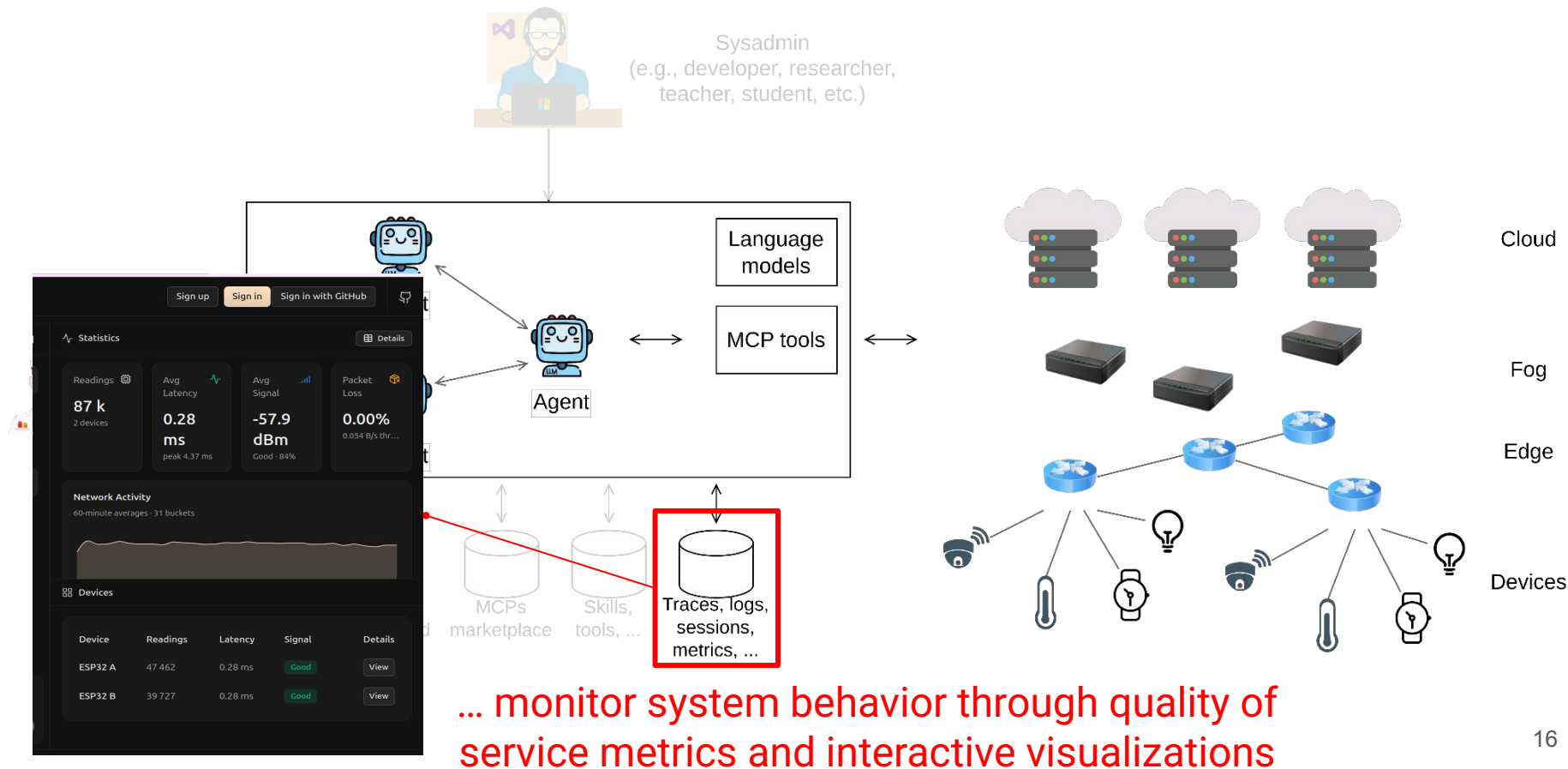


LLMThings: High-Level Architecture

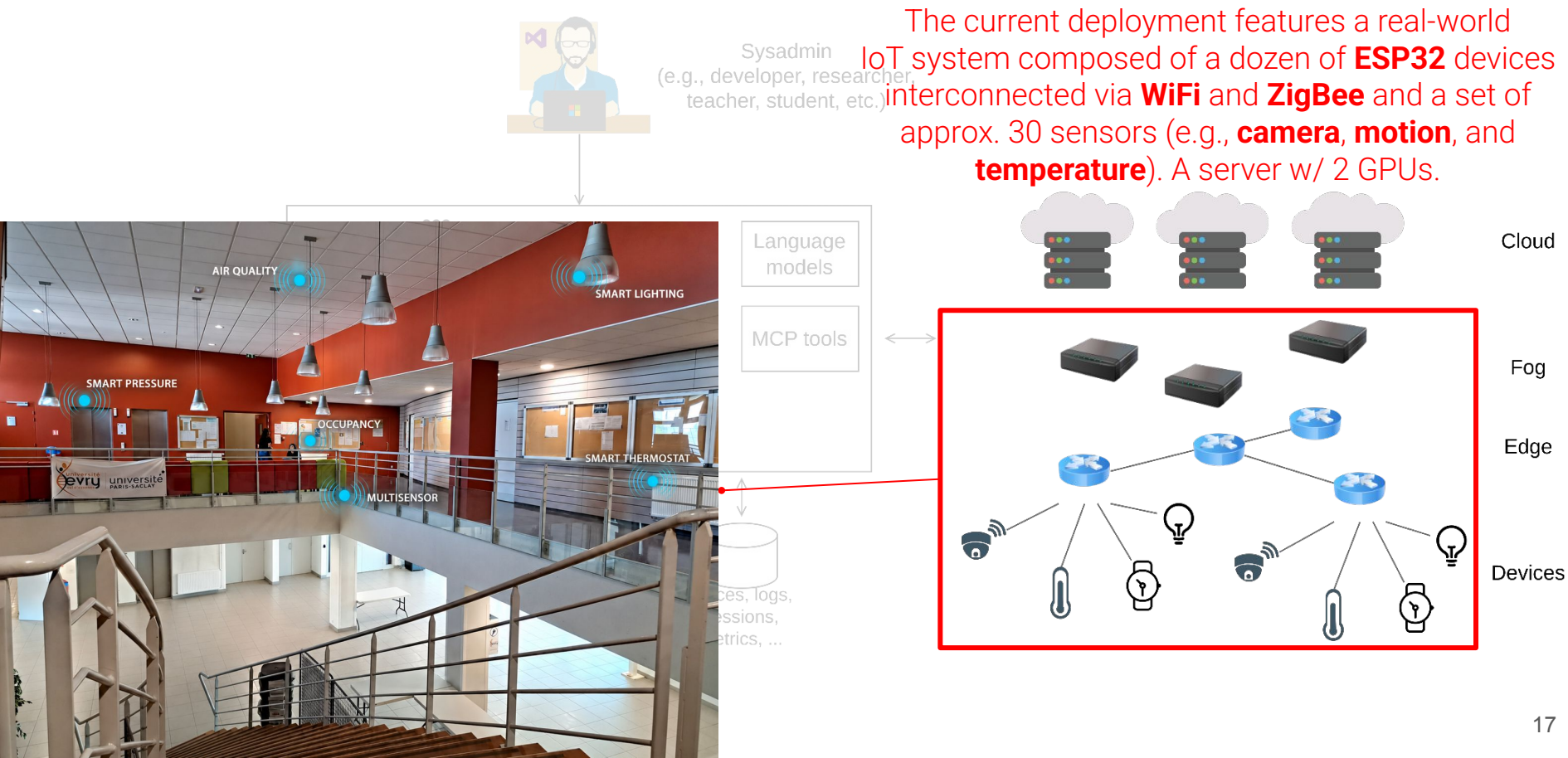


The workflow is executed in real-time on the IoT deployment

LLMThings: High-Level Architecture



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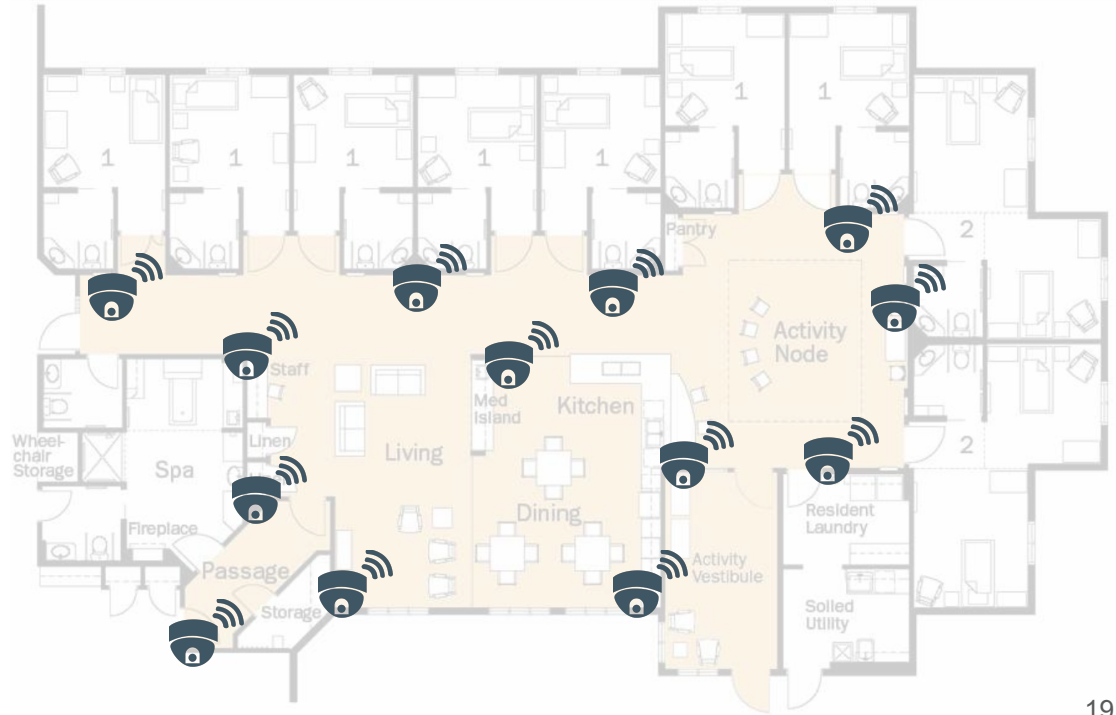
Fall prevention in a nursing home

- Nursing home arrangement replicated at IBGBI@Evry Univ.



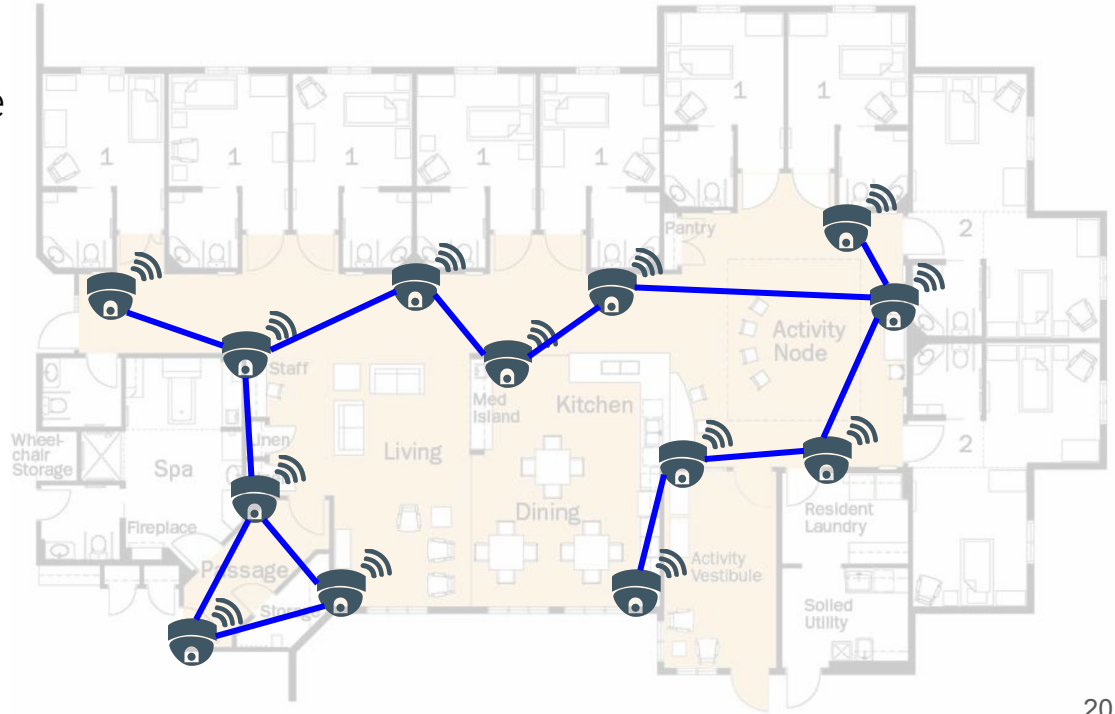
Fall prevention in a nursing home

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- IoT deployment
 - 13 Camera sensors spread evenly



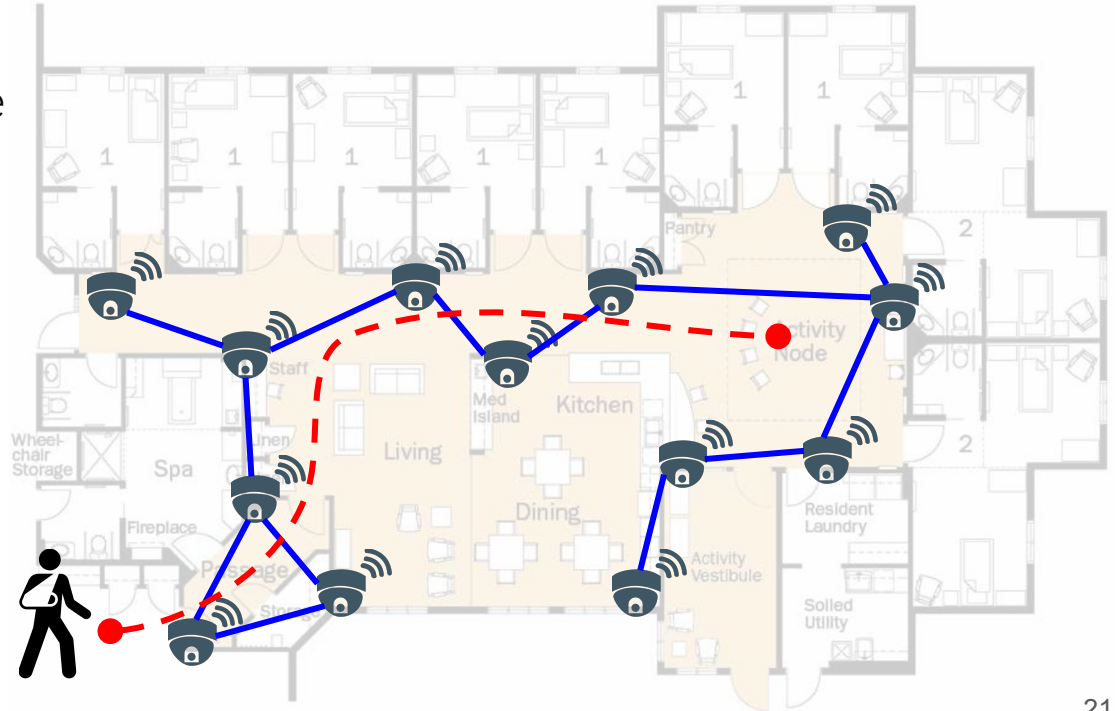
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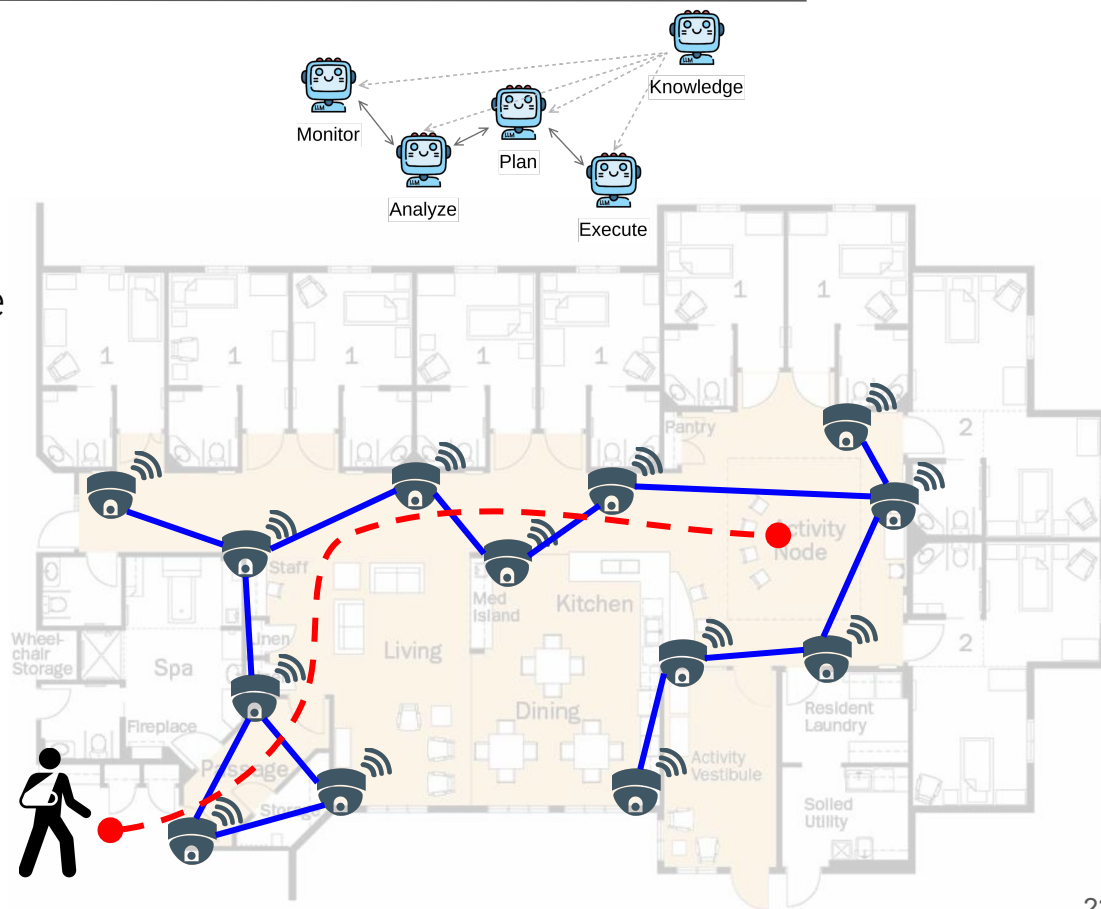
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 - track patient and ensure safety



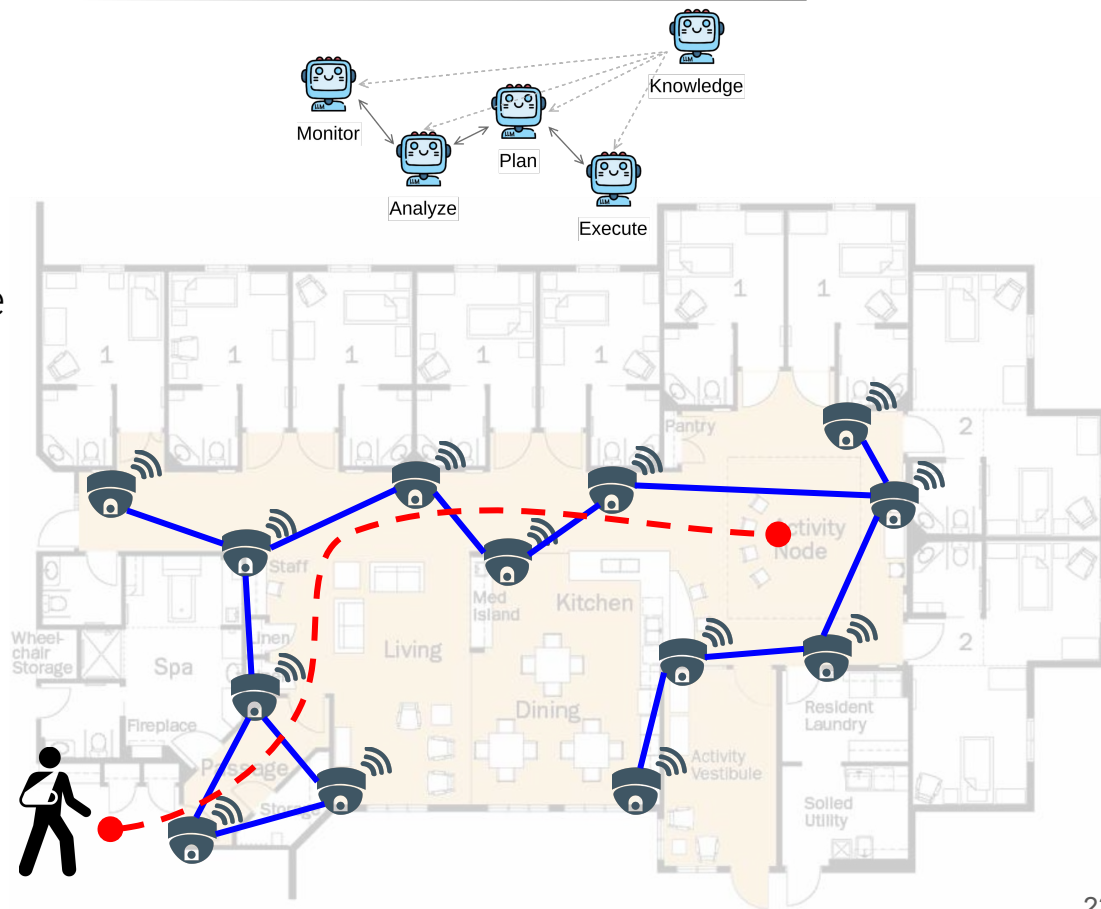
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- Orchestration
 - MAPE-K [12]
 - 5 Agents
 - Sequential coordination



Fall prevention in a nursing home

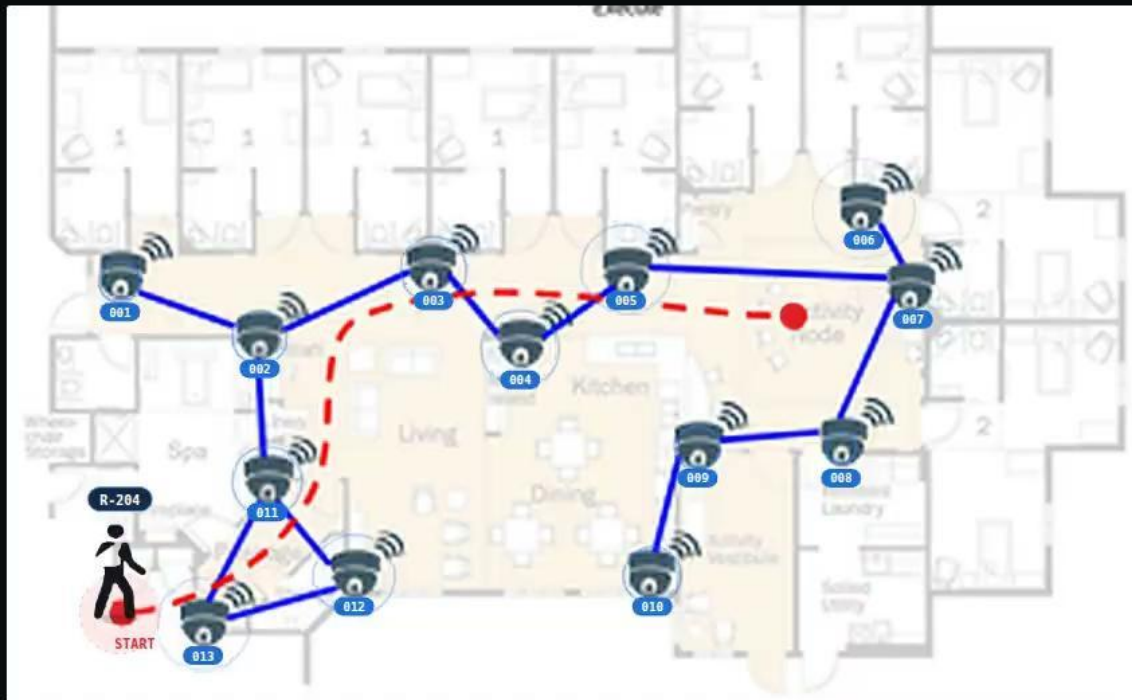
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- Scenario
 - fall prevention
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- Orchestration
 - MAPE-K [12]
 - 5 Agents
 - Sequential coordination
- Constraints
 - save energy
 - optimize bandwidth



Fall prevention in a nursing home ... In action

ESP32-CAM MESH Resident tracking · smart nursing home

mesh online 14:32:08.03



DETECTION STREAM → MONITOR AGENT

0/7 events

LLMThings : a framework towards achieving the vision of autonomic computing in IoT systems using language agents

One platform, three audiences

Research

Benchmark orchestration strategies (e.g., MAPE-K, OODA, FOCAL...) on real IoT deployments.

Development

Plug in new agents, tools and devices through MCP.

Education

Already in use in the *Agentic IoT course* at Univ. Évry.; organization of hackathons.

Acknowledgment to:

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- Département Informatique@Evry Uni.;
- Direction de l'Offre de Formation DOFREVA@Evry Uni.

Try the demo



lmthings.github.io/demo/

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